

budget. Luckily, scalable multi-robot simulation programs have come to the rescue.

Robotics: Real robots vs simulations

There exists a wide variety of robot simulators. Those designed before the year 2000 were typically targeted at specific use-cases. It was only during the last decade that PCs started to become powerful enough to allow for more complex designs. Today, there are a few general robot simulators that meet the needs of the robotics community.

The performance of a robot simulator and its accuracy are typically the result of a trade-off between the number of robots and speed of simulation. Performance evaluation of *Stage* and *ARGoS* discussed in later sections illustrate this trade-off. When the aim is to simulate large teams of robots, this trade-off can become dramatic. For this reason, in the last few years, a few simulators have been designed to provide good performance for large teams of robots.

The *Stage* simulator

Stage is a part of the Player Project (Player/Stage/Gazebo) and is now also a part of ROS (Robot Operating System); there have been significant changes since version 3 to improve its simulation performance. Maybe the most important novelty is the support for multi-threaded execution.

Stage is supported in nearly all UNIX-based OSs and in Mac OS X. It is interesting to know that the name ‘Player’ and ‘Stage’ were inspired by the quote, “All the world’s a stage,” by William Shakespeare in *As You Like It*.

Stage provides the user with a two-dimensional graphic environment, which has facilities for a perspective camera, making it effectively a 2.5 dimensional simulator. *Stage* has facilities for modelling the robot and its sensors using simple scripts. Some in-built *Stage* controllers are the ‘wander’ controller, which enables a wandering kind of behaviour; the ‘lasernoise’ controller, which helps in studying the noise generated in the laser sensor of the robot; and the ‘pioneer_flocking’ controller, which enables flocking behaviour for a number of pioneer robots.

To implement the ‘pioneer_flocking’ controller, *cd* to *Stage-source/worlds* and run the world file with the following command:

```
stage pioneer_flocking.world
```

That will pop up a screen similar to the one in Figure 1.

Stage shines at simulating navigation- and sensing-based experiments. It can simulate one robot about 1000 times faster than real-time, and 1000 robots in about real-time. However, the kinematic physics models employed in *Stage* prevent one from executing experiments that involve pushing and pulling objects, as well as robot self-assembly.

To cope with this issue, the Player Project offers Gazebo, which is a three-dimensional dynamics simulator. In Gazebo,

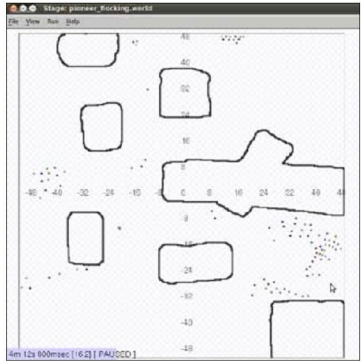


Figure 1: Multi-coloured pioneer robot models forming swarms

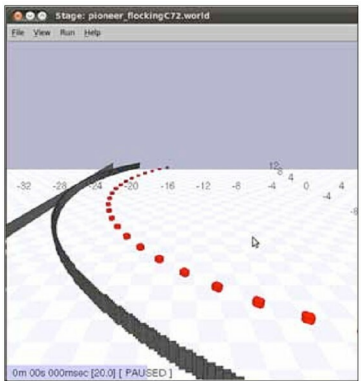


Figure 2: Pioneer robots formation

one can run very complex experiments. However, the improved accuracy of the robot models makes this simulator much slower than *Stage*.

Stage is also a part of ROS. However, at the time of writing, the latest ROS release *Electric Emys* does not support *Stage* controllers through the ROS nodes.

The stable release of *Stage* can be downloaded from playerstage.sourceforge.net/. Alternatively, if you feel adventurous and want to try the bleeding edge, you can download the development version at <https://github.com/rvtv/Stage>.